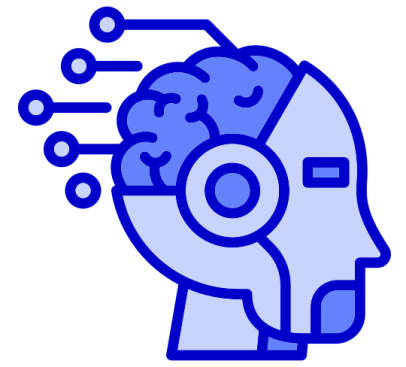
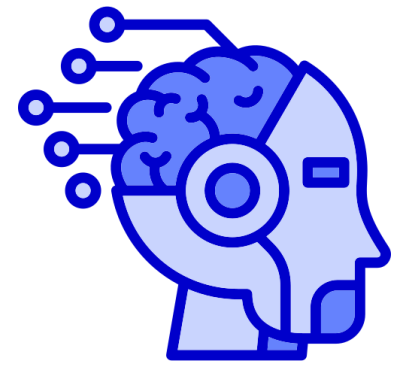




Machine Learning

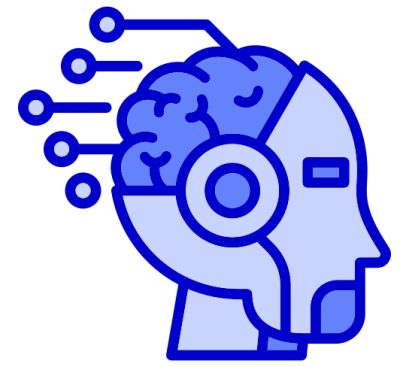
Linear Regression

Artificial Intelligence



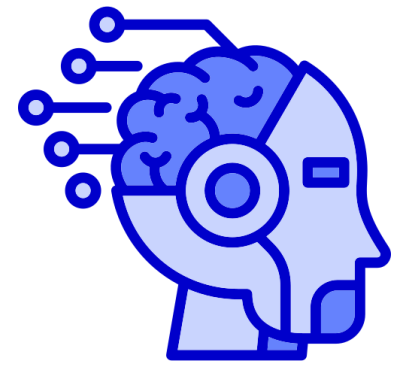
Outline

- Machine Learning
- Supervised Learning
 - Classification
 - Prediction
- Unsupervised Learning
 - Clustering
- Reinforcement Learning



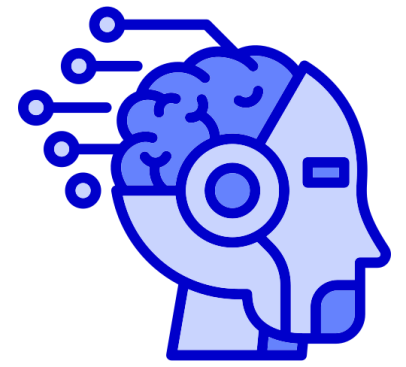
What is Learning

- Webster's definition of **to learn**
“To gain knowledge or understanding of, or skill in by study, instruction or experience”
 - Learning a set of new facts
 - Learning HOW to do something
 - Improving ability of something already learned



Why Learn?

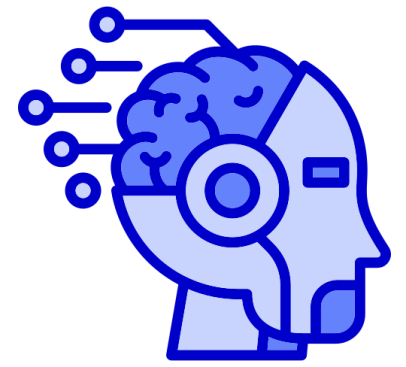
- There is no need to “learn” to calculate payroll
- **Learning is used when:**
 - Human expertise does not exist
 - Humans are unable to explain their expertise (speech recognition)
 - Solution changes in time (routing on a computer network)
 - Solution needs to be adapted to particular cases (user biometrics)



Learning

- Examples
 - Walking (motor skills)
 - Riding a bike (motor skills)
 - Telephone number (memorizing)
 - Playing backgammon (strategy)
 - Develop scientific theory (abstraction)
 - Language
 - Recognize fraudulent credit card transactions Etc.

(One) Definition of Learning

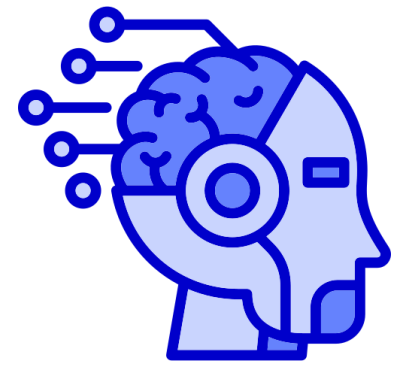


A computer program is said to learn from

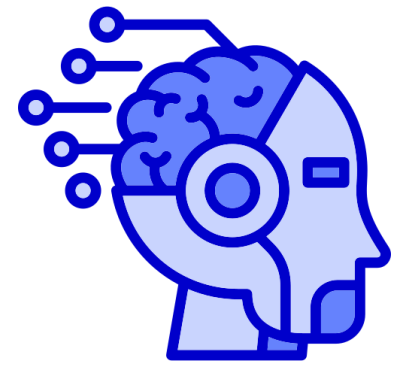
- experience E with respect to some class of
- tasks T and
- performance measure P ,

if its performance at tasks in T measured by P ,
improves with experience E .

Examples

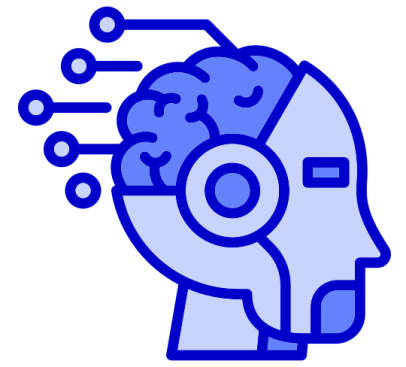


- Spam Filtering
 - T: Classify emails HAM / SPAM
 - E : Examples(e_1 , HAM), (e_2 , SPAM), (e_3 , HAM), (e_4 , SPAM)
 - P : Prob. of an error on new emails
- Personalized Retrieval
 - T: find documents the user wants for query
 - E : watch person use Google (queries/clicks)
 - P : # relevant docs in top10
- Play Checkers
 - T: Play checkers
 - E : games against self P: percentage wins



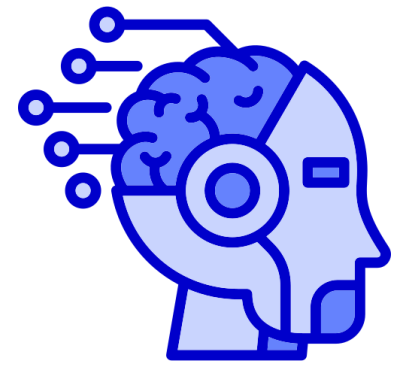
Supervised Learning

Artificial Intelligence



Supervised Learning

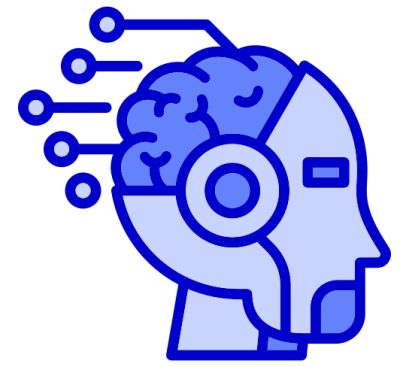
- Supervised learning is the machine learning task of inferring a function from **supervised training data**.
- The training data consist of a set of **training examples**.
- In supervised learning, each example is a pair consisting of an **input object** (typically a vector) and the desired **output value** (also called the supervisory signal).



Supervised Learning

A supervised learning algorithm analyzes the training data and produces

- (Inferred function) classifier
 - If the output is **discrete**. OR
- Regression function
 - If the output is **continuous**



Working of Supervised Learning algorithms

Given a set of training examples of the form:

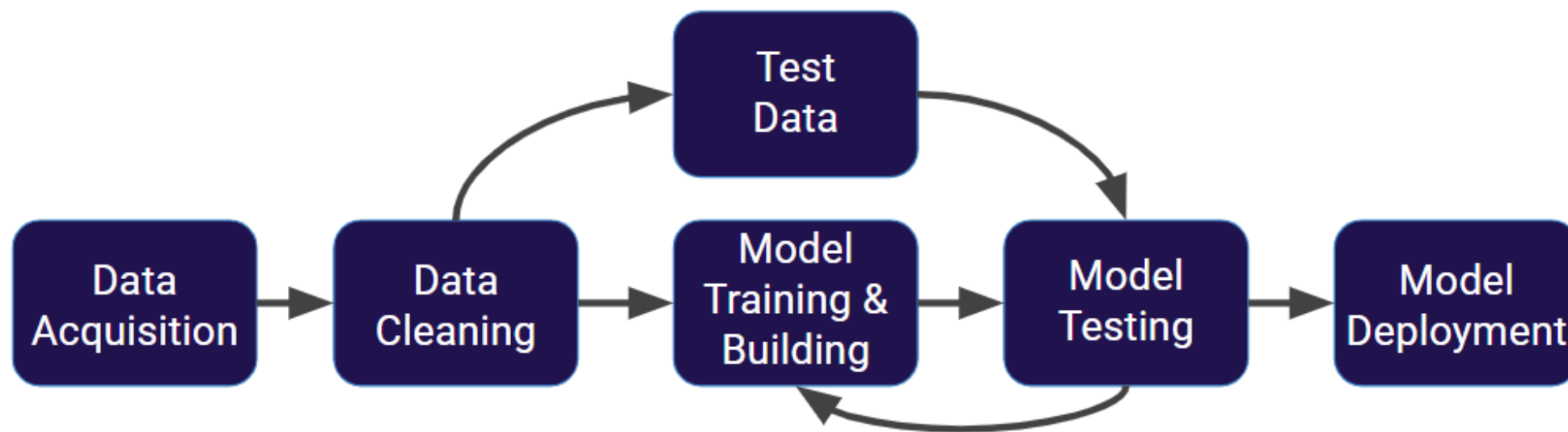
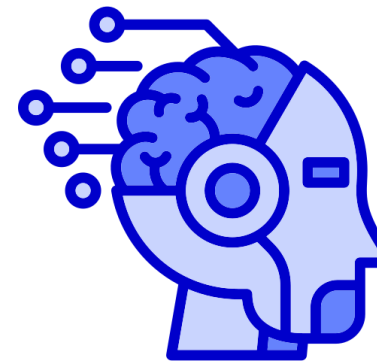
$$\{ (x_1, y_1), \dots, (x_N, y_N) \}$$

a learning algorithm seeks a function

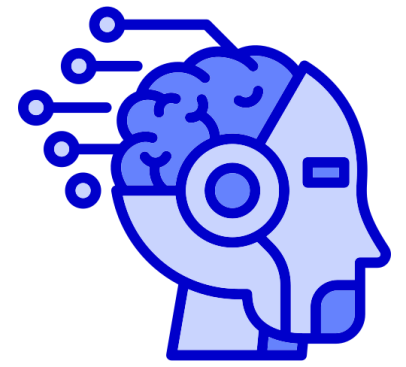
$$g : X \rightarrow Y$$

where X is the input space and Y is the output space and the function g is an element of some space of possible functions G , usually called the hypothesis space.

Machine Learning Process

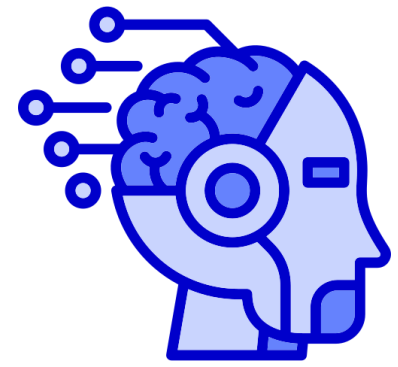


Artificial Intelligence



Classification

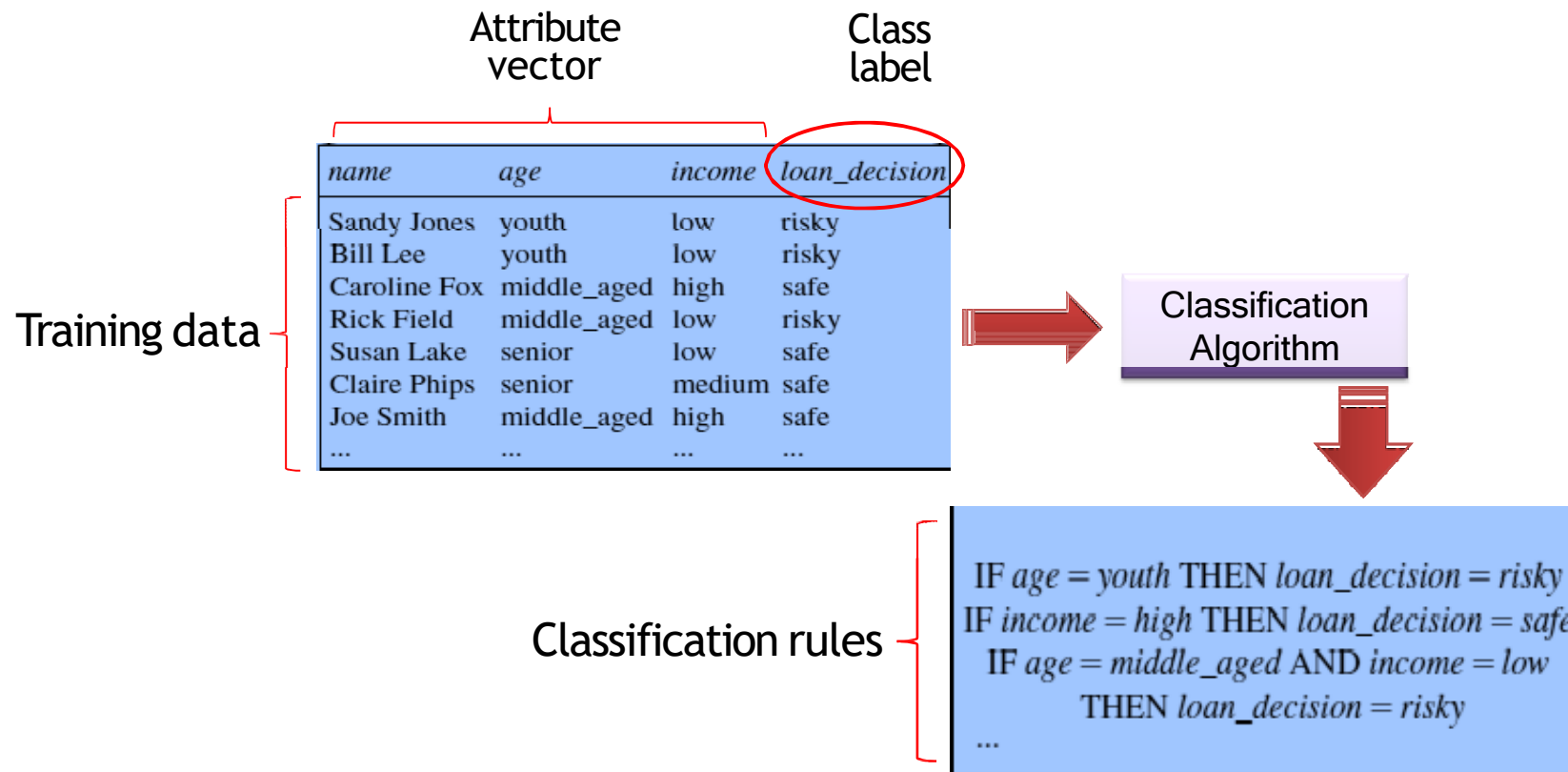
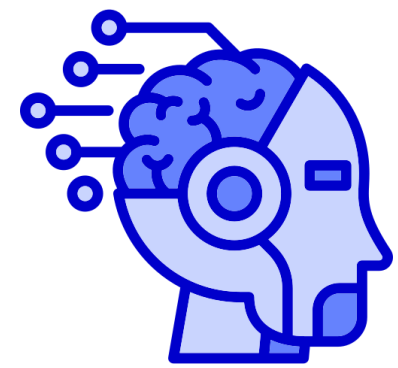
Artificial Intelligence

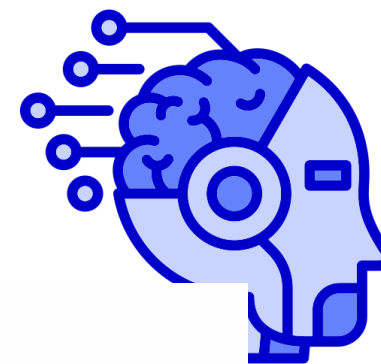


Classification: Definition

- Given a collection of records (*training set*)
- Each record contains a set of *attributes*, one of the attributes is the *class*.
- Find a *model* for class attribute as a function of the values of other attributes.
- **Goal**: previously unseen records should be assigned a class as accurately as possible.
- A *test set* is used to determine the accuracy of the model. Usually, the given data set is divided into training and test sets, with the training set used to build the model and the test set used to validate it.

The Basics - General Approach





Classification

Classification rules

IF *age* = *youth* THEN *loan_decision* = *risky*
IF *income* = *high* THEN *loan_decision* = *safe*
IF *age* = *middle_aged* AND *income* = *low*
THEN *loan_decision* = *risky*
...

Estimate classifier accuracy
(to avoid overfitting)

Predict classification
of new data

Test
data

<i>name</i>	<i>age</i>	<i>income</i>	<i>loan_decision</i>
Juan Bello	senior	low	safe
Sylvia Crest	middle_aged	low	risky
Anne Yee	middle_aged	high	safe
...	...	...	...

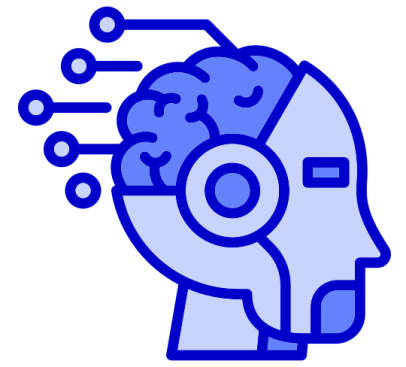
% test set tuples correctly classified

(Mervat Fahmy, youth, medium)
Loan decision?



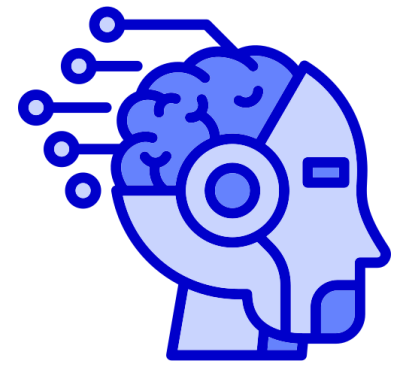
Risky

Artificial Intelligence



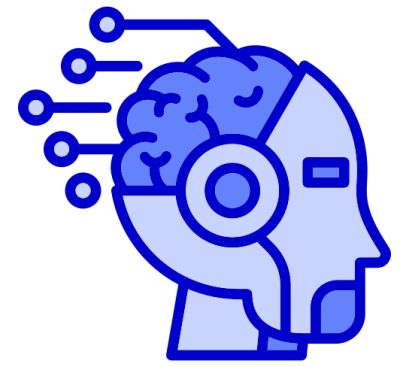
Examples of Classification Tasks

- **Predicting** tumor cells as benign or malignant
- **Classifying** credit card transactions as legitimate or fraudulent
- **Categorizing** news stories as finance, weather, entertainment, sports, etc



Different Types of Classifiers

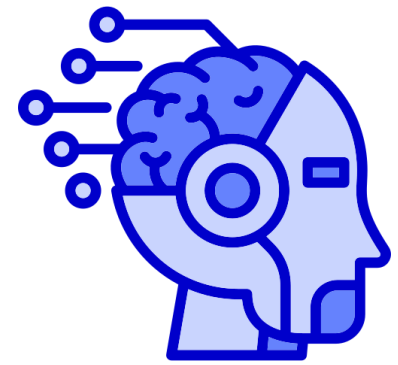
- Back propagation
- Bayesian Classifiers
- Decision Trees
- Density estimation methods
- Fuzzy set theory
- Linear discriminant analysis (LDA)
- Logistic regression
- Naive bayes classifier
- Nearest Neighborhood Classification
- Neural networks
- Quadratic discriminant analysis (QDA)
- Support Vector Machine
- many more...



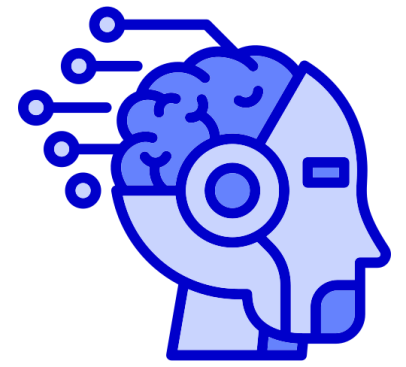
Prediction

Artificial Intelligence

What Is Prediction?

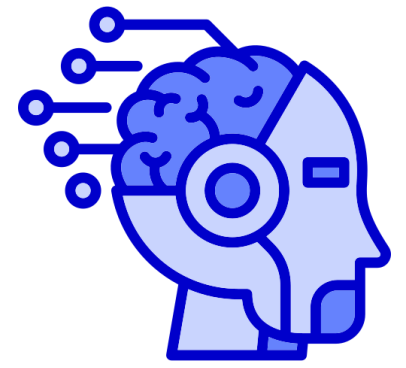


- Prediction is **similar** to classification
 - First, construct a model
 - Second, use a model to predict the unknown value
 - Major method for prediction is regression
 - Linear and multiple regression
 - Non-linear regression
- Prediction is **different** from classification
 - Classification refers to predicting the categorical class label
 - Prediction models continuous-valued functions



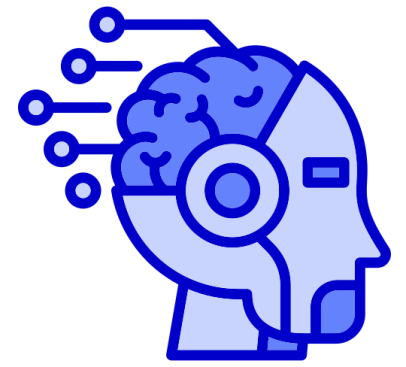
Unsupervised Learning

Artificial Intelligence



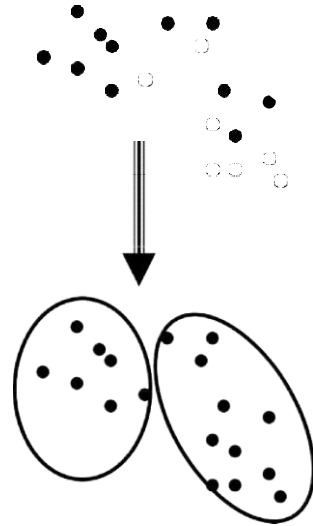
Unsupervised Learning

- Unsupervised Learning takes the set of attributes/features as training examples **alone**.
- The purpose of unsupervised learning is to attempt to find natural **partitions** in the **training set**.
- Two general strategies for Unsupervised learning include Clustering and Hierarchical Clustering.

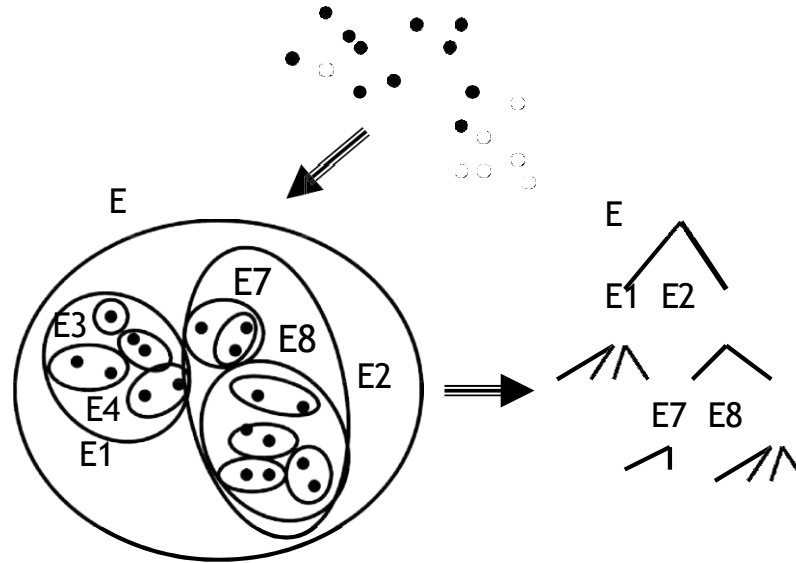


Clustering and Hierarchical Clustering

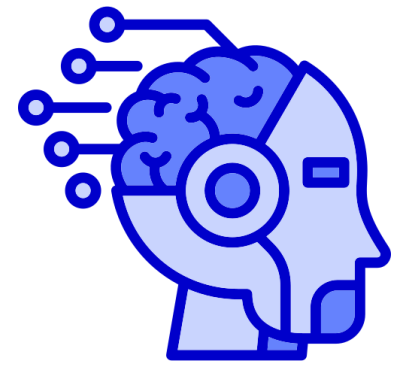
1. Clustering



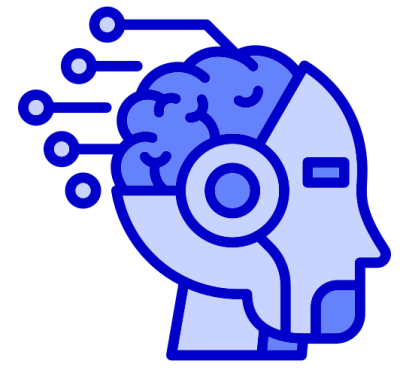
2. Hierarchical Clustering



What is Unsupervised Learning Useful for?



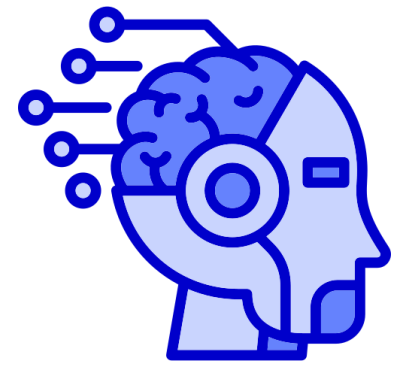
- Collecting and labeling a large set of sample patterns can be very expensive. By designing a basic **classifier with a small set of labeled samples**, and then tuning the classifier up by allowing it to run without supervision on a large, unlabeled set, much **time and trouble can be saved**.
- Training with **large amounts** of data often less expensive, unlabeled data, and then **using supervision to label the groupings found**. This may be used for large "data mining" applications where the contents of a large database are not known beforehand.



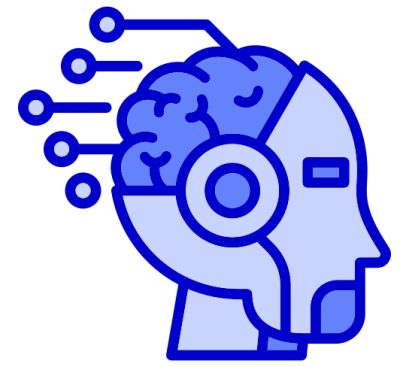
What is Unsupervised Learning Useful for?

- Unsupervised methods can also be used to find **features** that can be useful for **categorization**. There are unsupervised methods that represent a form of data-dependent "smart pre-processing" or "smart feature extraction."
- Lastly, it can be of interest to gain **insight** into the **nature** or **structure** of the **data**. The discovery of **similarities** among patterns or of major **departures** from expected characteristics may suggest a significantly different approach to designing the classifier.

Other Unsupervised Methods



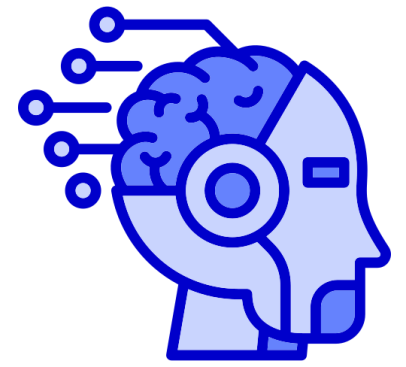
- There are a lot of other Unsupervised Learning Methods.
- **Examples:**
 - k-means
 - The EM Algorithm
 - Competitive Learning
 - Kohonen's Neural Networks: Self-Organizing Maps
 - Principal Component Analysis, Autoassociation



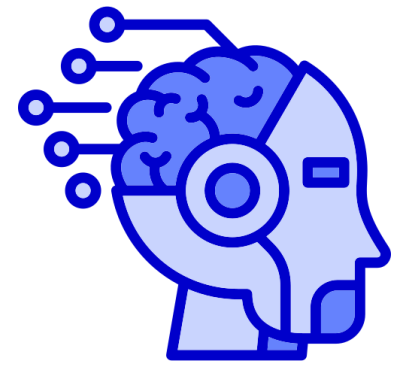
Reinforcement Learning

Artificial Intelligence

Reinforcement learning



- Intelligent learning algorithm
- Doesn't require the presence of a teacher
- The algorithm is given a reward (a reinforcement) for good actions
- The algorithm tries to figure out what is the best action to take in a given state, without knowing the final optimal solution.
- The actions are based on rewards and penalties.



Applications

- Robot control
- Elevator scheduling (search for patterns)
- Telecommunications (finding networks)
- Games (Chess, Backgammon)
- Financial trading